DAY-2

TCP/IP Suite Theory (1 hour): What is the TCP/IP Suite? Layers of TCP/IP: Application, Transport, Internet, and Network Access. Practical (1 hour): Analyze packet headers to identify the TCP/IP layer using Wireshark. Task: Compare OSI and TCP/IP models with real-world examples

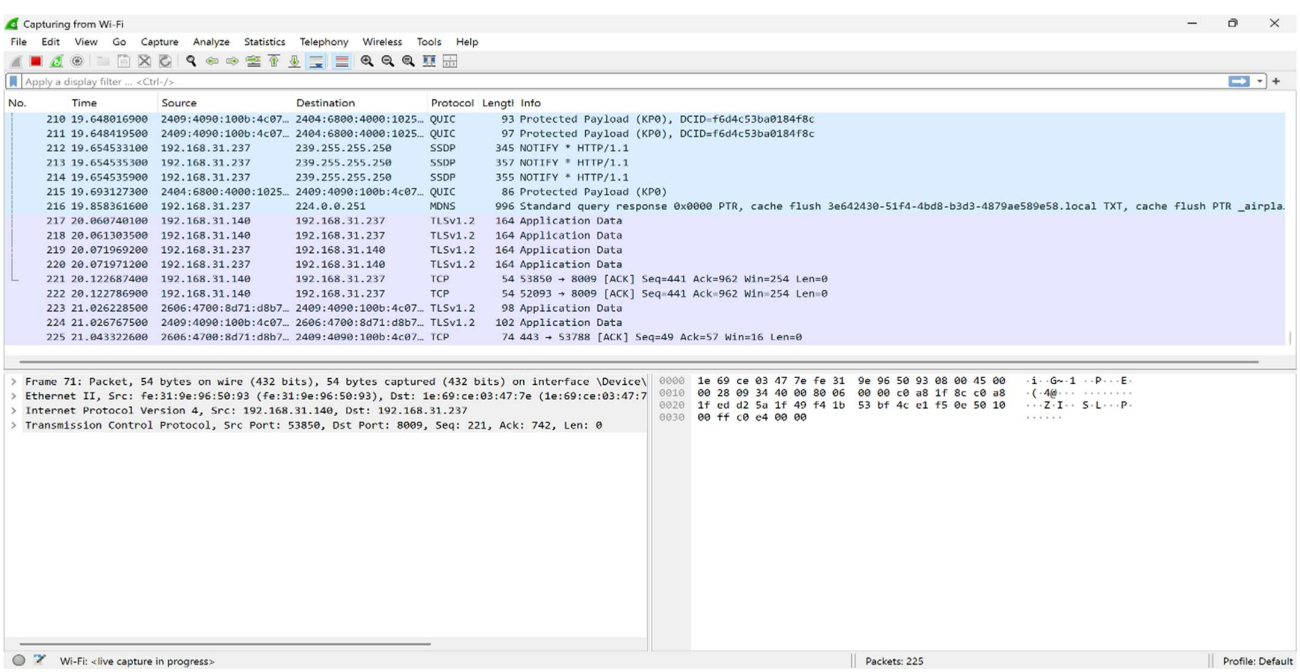
1. Wireshark Packet Capture Analysis

Open Wireshark and start capturing packets. Refresh the Flipkart homepage and observe the captured traffic.

TCP/IP Layer	Example Packet	How Identified
Application Layer	HTTP/HTTPS Packet	Protocol column shows HTTP or TLS/HTTPS traffic
Transport Layer	TCP Segment	Contains Source Port, Destination Port, Sequence Number
Internet Layer	IP Packet	Contains Source IP and Destination IP addresses
Network Access Layer	Ethernet Frame	Contains Source MAC and Destination MAC addresses

Example Observation

- **Application Layer:** HTTPS request sent to Flipkart server.
- **Transport Layer:** TCP packet using port 443.
- **Internet Layer:** IPv4 packet containing source and destination IP addresses.
- **Network Access Layer:** Ethernet frame carrying the packet over the local network.



2. Comparison of OSI and TCP/IP Models

OSI Model Layer	TCP/IP Model Layer	Real-World Example
Application	Application	Sending a WhatsApp message
Presentation	Application	Data encryption in WhatsApp
Session	Application	Maintaining an Instagram login session
Transport	Transport	Reliable delivery of a Zomato order request using TCP
Network	Internet	IP routing to Flipkart servers
Data Link	Network Access	Ethernet or Wi-Fi communication
Physical	Network Access	Electrical signals or wireless radio waves

Example

When ordering food on Zomato:

- Application Layer → Creates the order request.
- Transport Layer → Breaks data into segments using TCP.
- Internet Layer → Adds IP addresses for routing.
- Network Access Layer → Sends data over Wi-Fi or Ethernet.

3. TCP Segment Header Fields

Field	Function
Source Port	Identifies the sending application or service.
Destination Port	Identifies the receiving application or service.
Sequence Number	Keeps track of the order of transmitted data.
Acknowledgment Number	Confirms receipt of data from the sender.
Header Length	Indicates the size of the TCP header.
Flags	Control connection setup and termination (SYN, ACK, FIN, etc.).
Window Size	Specifies how much data can be received before acknowledgment.
Checksum	Detects errors in the TCP segment.
Urgent Pointer	Indicates urgent data that requires immediate processing.

Examples of Common Flags

- SYN → Starts a connection.
- ACK → Acknowledges received data.
- FIN → Ends a connection.
- RST → Resets a connection.

4. Message Flow Through TCP/IP Layers

```
User Sends Instagram Message
        │
        ▼
+----------------------+
| Application Layer    |
| Message Created      |
+----------------------+
        │
        ▼
+----------------------+
| Transport Layer      |
| Message Split into   |
| TCP Segments         |
+----------------------+
        │
        ▼
+----------------------+
| Internet Layer       |
| IP Address Added     |
+----------------------+
        │
        ▼
+----------------------+
| Network Access Layer |
| Ethernet/Wi-Fi Frame |
| Created and Sent     |
+----------------------+
        │
        ▼
Instagram Server
```

Layer Functions

- Application Layer → Creates and processes the message.
- Transport Layer → Splits data into packets and ensures reliable delivery.
- Internet Layer → Adds source and destination IP addresses.
- Network Access Layer → Transmits data through Wi-Fi or Ethernet.

5. Analysis of One Captured Packet

Selected Packet

HTTPS Packet (TCP Port 443)

TCP/IP Layer

Application Layer

How It Was Identified

- Protocol column displayed **TLS/HTTPS**.
- Destination port was **443**, which is commonly used for secure web traffic.

Real-World Action Represented

This packet could represent:

- Loading the Flipkart homepage.
- Sending a WhatsApp message.
- Booking a ticket on BookMyShow.
- Logging into Instagram.

The packet carries encrypted application data between the user device and the server using HTTPS.